Class | Adm No

Candidate Name: ____________________________

2016 Preliminary Examination II
Pre-University 3

H2 Biology **9648/03**

Applications Paper and Planning Question **22 September 2016**

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.
You will lose marks if you do not show your working or if you do not use appropriate units.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This question paper consists of 21 printed pages including 1 blank page

[Turn over

For Examiner's Use

Answer **all** questions.

1. Before the emergence of more advanced technologies, cDNA libraries have been used to study the genetic changes involved in cancer development.

Normal and tumour cells are obtained from the same patient. Reverse transcription is carried out on mRNA isolated from the tumour cells. Primers consisting of thymine repeats were used during reverse transcription to form single-stranded cDNA as depicted in Figure 1.1. These cDNA are then labelled with fluorescent dyes.

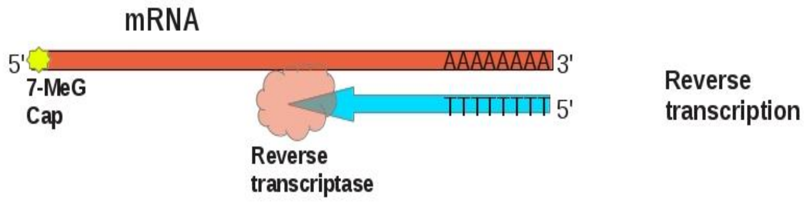

Figure 1.1

Normal cell mRNA and tumour cell cDNA are allowed to hybridise; the resulting double-stranded hybrid molecules and remaining single-stranded mRNA are discarded. Subtracted cDNA (also known as non-hybridized cDNA) are used to form a subtracted cDNA library. The process is summarized by Figure 1.2.

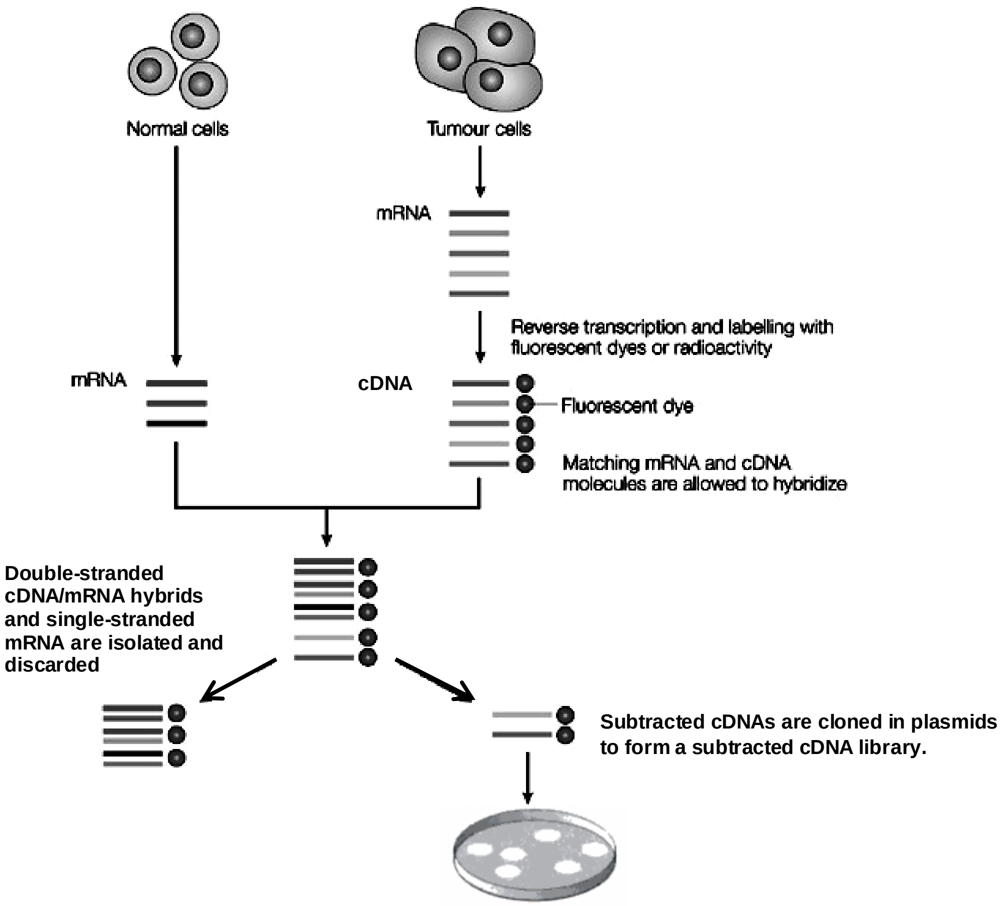

Figure 1.2

For Examiner's Use

A young scientist conducting this procedure was concerned that possible presence of bacteria within cell samples might result in production of bacteria cDNA. This would contaminate downstream processes. However, her concerns were dismissed by her partners who assured her that reverse transcription of bacterial mRNA is unlikely to occur.

(a) Using the information given, explain why reverse transcription of bacterial mRNA is unlikely to occur.

Bacterial mRNA do <u>not possess a 3' poly(A) tail</u> ;
This because bacteria <u>does not carry out post-transcriptional modification</u> to its mRNA.
Hence oligonucleotide <u>primer is unlikely to anneal</u> to bacterial mRNA for reverse transcription;
Max 2

2]

(b) Describe the steps needed to create recombinant DNA molecules that are used to assemble a plasmid library of subtracted cDNA.

<u>Complementary strand of cDNA</u> with respect to subtracted cDNA sequences are synthesized <u>using DNA polymerase</u>;
<u>Linkers</u> containing appropriate restriction sites are <u>ligated to the double stranded cDNA;</u>
cDNA and plasmids are <u>cut/digest</u> using the same restriction enzyme to produce <u>complementary sticky ends;</u>
Digested cDNA are <u>ligated</u> with digested plasmids;
During ligation digested cDNA and plasmids anneal with each other by <u>complementary base pairing / Hydrogen bonding</u> and are joined using <u>DNA ligase</u> which catalyses the <u>formation of phosphodiester bonds;</u>
Max 4

[4]

(c) Explain why the DNA sequences in the subtracted cDNA library are considered mutant alleles implicated in cancer.

cDNA from tumour cells that hybridizes with mRNA from normal cells represent alleles that are <u>expressed in both</u> tumour and normal cells;
Alleles that are expressed in both tumour and normal cells <u>may not be implicated in cancer;</u>
Alleles that are expressed in both tumour and normal cells are removed by discarding double stranded cDNA/mRNA hybrids;
Subtracted cDNA derived from tumour cells <u>represents the alleles that are expressed in tumour cells but not in normal cells;</u>
Max 2

]

(d) Suggest why the subtracted cDNA library may not fully capture all possible DNA sequences implicated in cancer.

Loss- of-function <u>mutations in promotor sequence of tumour suppressor genes</u> would not be captured by subtracted cDNA library;
Loss- of-function mutations in promotor sequence would <u>inhibit gene expression;</u>
AVP
Max 1;

1]

Screening of the subtracted cDNA library of a cancer patient revealed a mutated protein kinase gene. A research team decided to clone this gene to isolate the mutant protein and study it to better understand its role in cancer development.

The protein kinase gene was first isolated. The artificial plasmid, pKY350, was constructed to act as an expression vector.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, *BamHI*, in the middle of the tetracycline resistance gene.

A pKY350 plasmid was cut using *BamHI* and the cDNA sequence for the mutant protein kinase was inserted into it. Figure 1.3 shows pKY350 and the recombinant plasmid.

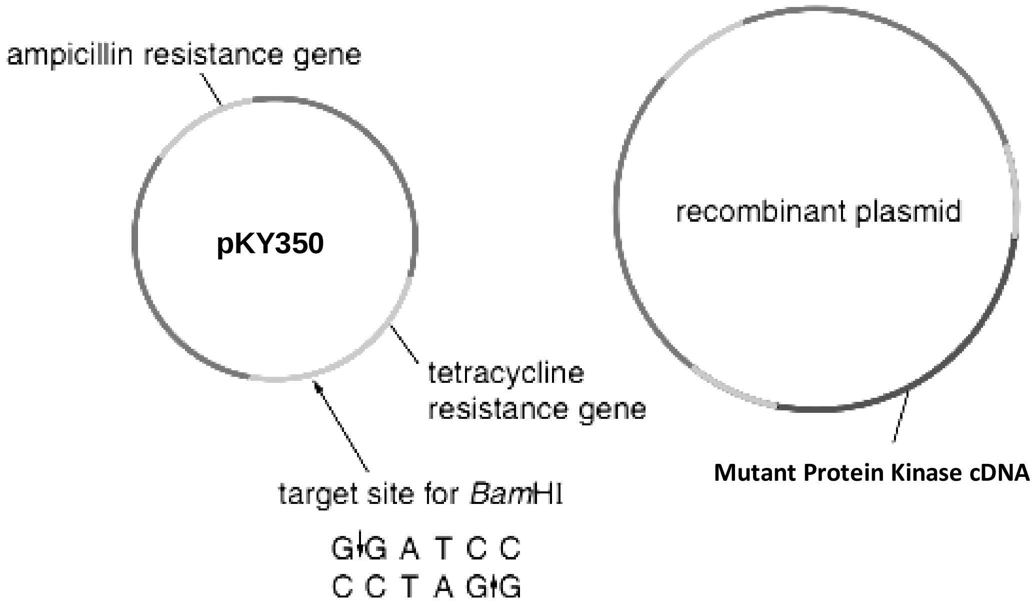

Figure 1.3

The mixture containing recombinant DNA was used to transform *E.coli* bacteria. Replica plating was used to identify recombinant bacteria with the mutant protein kinase gene. Figure 1.4 shows the bacterial colonies that grew on different nutrient agar plates.

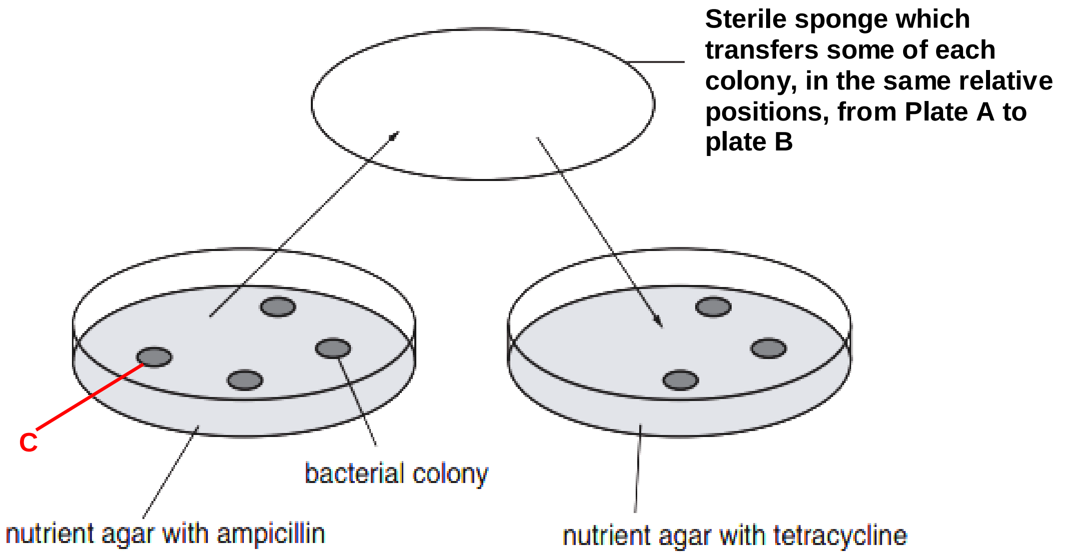

Plate A after incubation **Plate B after incubation**

Figure 1.4

(e) Use a label line and the letter C to identify, on Figure 1.4, a colony of bacteria that contain the recombinant plasmid.

[1]

A senior scientist declared that it would be a challenge to develop a good understanding of the role of the mutant protein kinase in cancer development from studying the recombinant protein produced by a bacterial cloning system.

(f) Explain why recombinant mutant protein kinase produced by a bacterial cloning system may not be a good model of study for understanding cancer deve

Mutant tyrosine kinase synthesized in bacteria cells will <u>not undergo post-translational modifications</u>;
Therefore the mutant protein may <u>not have the same structure</u> as it would adopt in normal cancer cells;
Mutant protein kinase may <u>function differently</u> with a different structure which would undermine inferences about its role in cancer cells;
Max 2m

.... [2]

Today, due to the findings of the human genome project, polymerase chain reaction (PCR) has largely replaced cDNA libraries to directly isolate genes in the study of diseases. PCR is normally carried out using the enzyme Taq DNA polymerase. This enzyme was originally extracted from the bacterium *Thermus aquaticus*. This bacterium was found in Mushroom Spring, one of the hot springs in Yellowstone National Park in the USA.

Taq DNA polymerase is now obtained from genetically modified *Escherichia coli* that carry the Taq DNA polymerase gene. The optimum temperature of this enzyme is 75-80°C.

After 40 minutes at 95°C, the activity of the enzyme is reduced by half. This means that the half-life of this enzyme at 95°C is 40 minutes.

(g) Using your knowledge of PCR, explain why a half-life of 40 minutes at 95°C allows many cycles of PCR before the enzyme needs to be replaced.

Each cycle of PCR consists of the 3 stages of denaturation, annealing and extension;
Only the denaturation stage occurs at 95^0C and annealing and extension occur at lower temperatures of 54^0C and 72^0C respectively;
Each cycle is completed within a short period of time which allows for many cycles of PCR to occur in 40min.
Taq DNA polymerase has a high optimum temperature which results in a long half-life of 40 minutes;

Max 3

[3]

(h) Suggest why Taq DNA polymerase is now obtained from genetically modified *E.coli.*

Cloning with E.coli allows for large scale production of Taq DNA polymerases;
It is inconvenient to recover *Thermus aquaticus* to maintain constant supply of natural Taq DNA polymerases;
Thermus aquaticus is not easily cultured under laboratory conditions;
AVP
Max 1

[2]

Work on the human genome project was able to provide many benefits to the field of molecular medicine.

(i) State 2 other areas that have benefited from work on the human genome project.

Study of Microbial Genomics;
Risk Assessment;
Study of Bioarcheology, Anthropology, Evolution, and Human Migration;
Genomic Mapping;
Max 2

[2]

[Total: 19]

2. Pluripotent stem cells are one type of stem cells and there is widespread interest in their study today. One key reason is because pluripotent stem cells can be induced to differentiate into the specific cell type required to repair damaged or destroyed cells or tissues. Additionally pluripotent stem cells can be used to study early events in human development and find out more about how cells differentiate and function. This may help researchers find out why some cells become cancerous and how some genetic diseases develop.

In 2006, a Japanese Scientist, Shinya Yamanaka, made a ground-breaking finding that would win him the Nobel Prize in Physiology or Medicine just six years later. He discovered that specialized cells can be stimulated to dedifferentiate and change back into pluripotent stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In experiments with mice, Yamanaka showed that the introduction of four genes caused specialized cells to change to iPS cells. Further studies have shown that these four genes indirectly result in the reprogramming of specialized cells by influencing the activity of many other genes.

(a) Outline how the addition of the four genes may influence the activity of many other genes to result in the reprogramming of specialized cells into iPS cells.

The genes code for transcription factors / activators and repressors respectively/chromatin remodelling enzymes;
These proteins mediate changes in gene expression to result in dedifferentiation into pluripotent stem cells;
These proteins act to increases the expression of genes that control pluripotency of cells;
These proteins act to decreases the expression of genes that resulted in differentiation of cells;
Max 2

[2]

For Examiner's Use

Pluripotent stem cells can also be derived from cytoplasmic hybrid (cybrid) cells through the method of somatic cell nuclear transfer (SCNT). A procedure for producing pluripotent stem cells by SCNT is outlined in Figure 2.1.

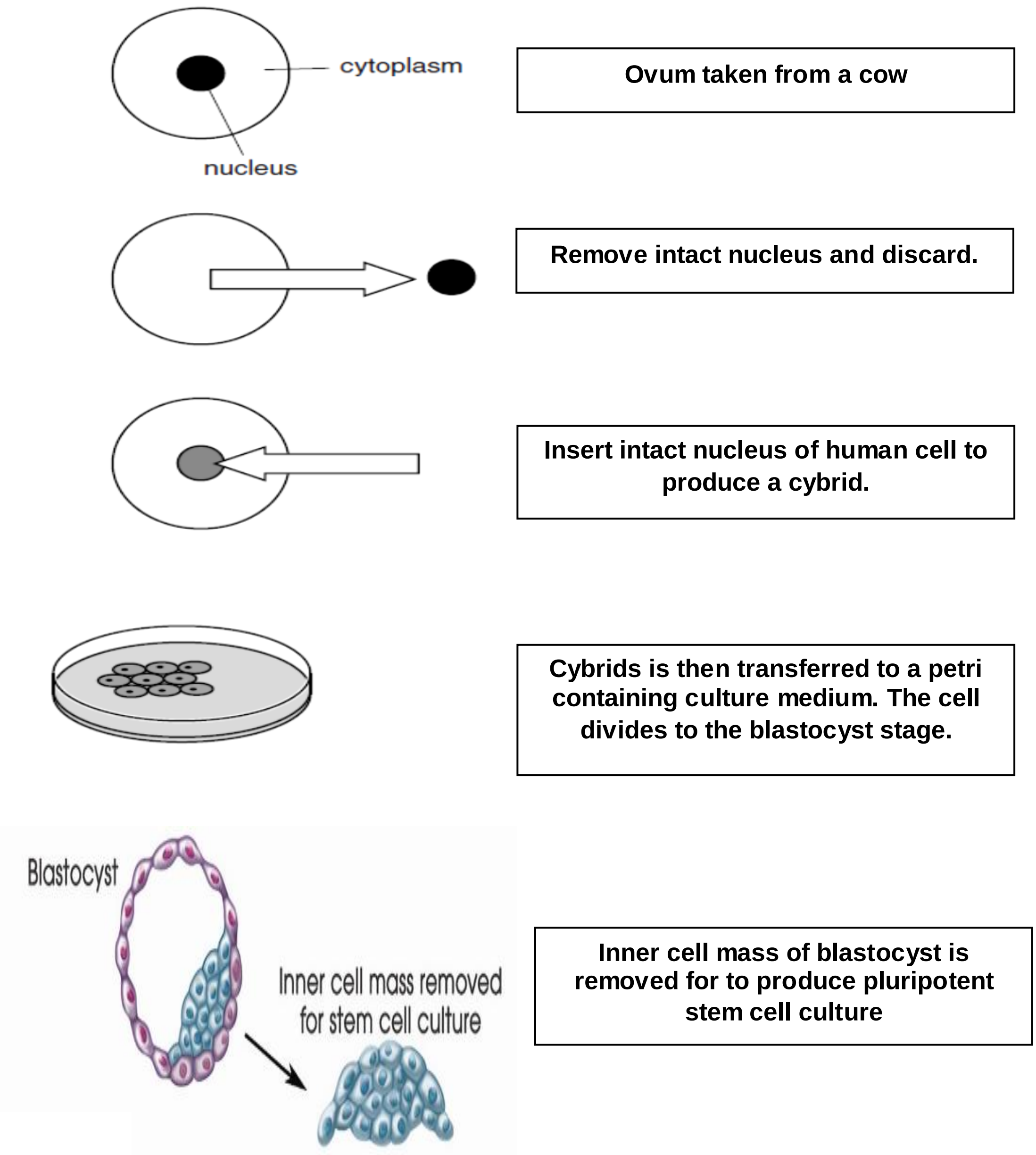

Figure 2.1

The DNA from the pluripotent stem cells generated by the above method is found to be 99.6% human.

(b) Explain the origins of the 0.04% cow DNA that is found in these cells.

Mitochondria from cow's ovum;
Binary division of mitochondria as cells divides account for presence in descendent pluripotent cells;

[2]

Some people have argued that it is unethical to allow the production of cybrids.

(c) Suggest why the production of hybrids may be considered unethical.

Presence of human and animal DNA means it is partly human and partly cow/ a human-animal hybrid which would not happen in nature;
Claims of the benefits of embryonic stem cell research are overrated/ few examples of success of using embryonic stem cells in medical field;
May lead to abuse of technology in future;
Unnecessary because, there already are alternative techniques to generate pluripotent stem cells such as by iPS technology;
Adult stem cells can be used to treat diseases;
AVP

Reject idea that human embryos will be destroyed

In 2008 … her ow… were u… her tra… multipli… replace the damaged section of her bronchus. A month later the transplanted tissue had developed its own blood supply. This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

(d) State two reasons why using the patient's own stem cells to treat a damaged bronchus is preferred to a transplant involving another donor.

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……………………………………………………………………………………………

…………………………………………………………………………………………… [2]

There will <u>not be any immune response / tissue rejection</u> since no foreign antigens, foreign tissue from the donated bronchus;
Overcomes the problem of <u>finding and locating suitable donors</u>;
No risk of infection arising from donor tissue / no need to decide who is eligible;
No need for immunosuppressant drugs to avoid rejection of transplanted organ;
Bronchus cells can be obtained / <u>cultured in large quantities</u> from the patient's stem cells;
Healthy culture of patient's stem cells can be <u>stored for future needs</u>;
AVP
Max 2

(e)

……………………………………………………………………………………………

……………………………………………………………………………………………

…………………………………………………………………………………………… [3]

<u>Multipotent</u> stems;
<u>Partially differentiated;</u>
Can differentiate further to produce cells within the same family;
Adult stem cells are <u>self-renewing</u> and can go through many rounds of cell division without differentiating;
AVP
Max 3

3. Iron-deficient anaemia is one of the most serious problems worldwide. Some scientists decided to explore the use of transgenic rice with increased copies of the Nicotianamine Synthase (NAS) gene to address the problem of iron deficiency. NAS is an enzyme in the metabolic pathway involved with iron acquisition in the rice plant. In these regards, the use of transgenic rice to address human iron deficiency is based on developing rice with increased iron content as transgenic rice would contain more copies of the NAS gene and produce more NAS enzymes compared to wild type rice.

To produce transgenic rice, a recombinant Ti plasmid containing NAS gene is engineered and reintroduced back into the bacteria *Agrobacterium tumeficiens*. The recombinant bacteria is then allowed to infect the plant. The process showing how transgenic rice plant is produced is illustrated in Figure 3.1.

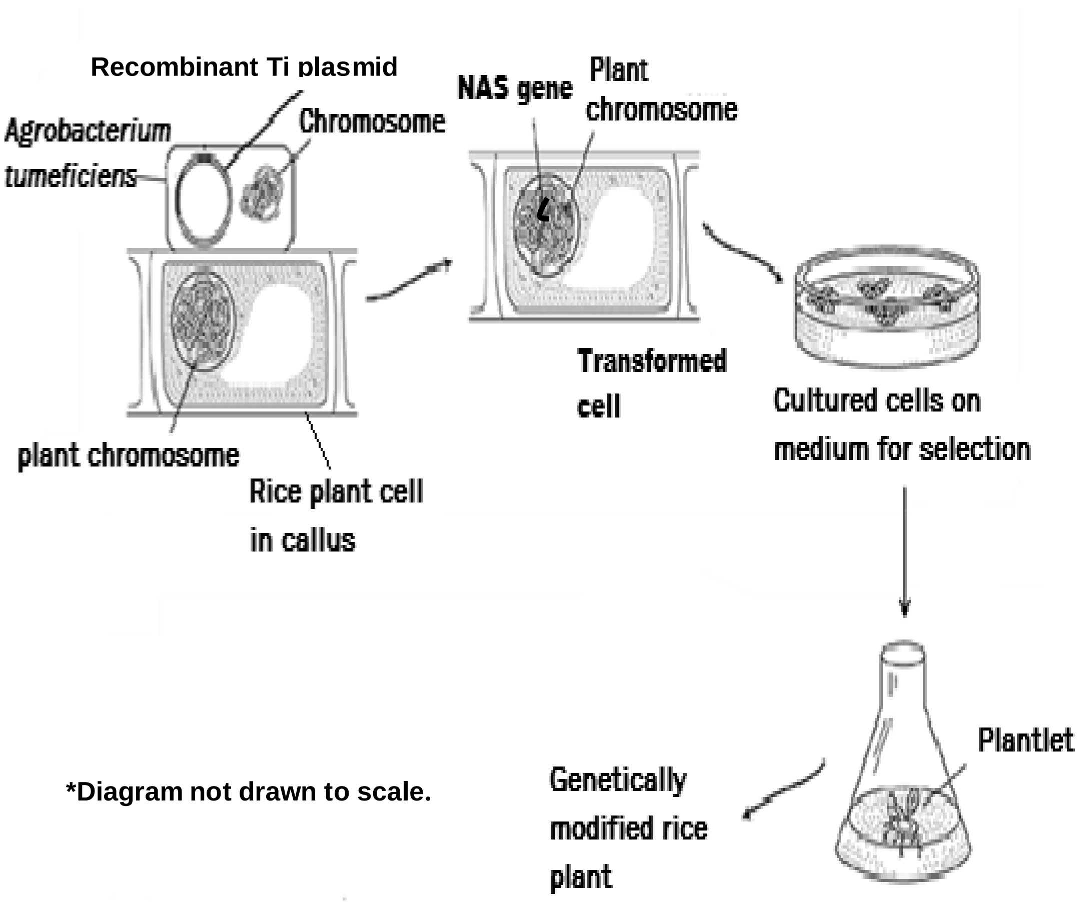

Figure 3.1

The naturally occurring Ti plasmid derived from *Agrobacterium tumefaciens*, is commonly used as a vector for introducing new foreign genes into plant cells. This plasmid integrates a segment of its DNA known as T-DNA, into the chromosomal DNA of host plant cells after infection. Figure 3.2 shows the structure of a natural Ti plasmid.

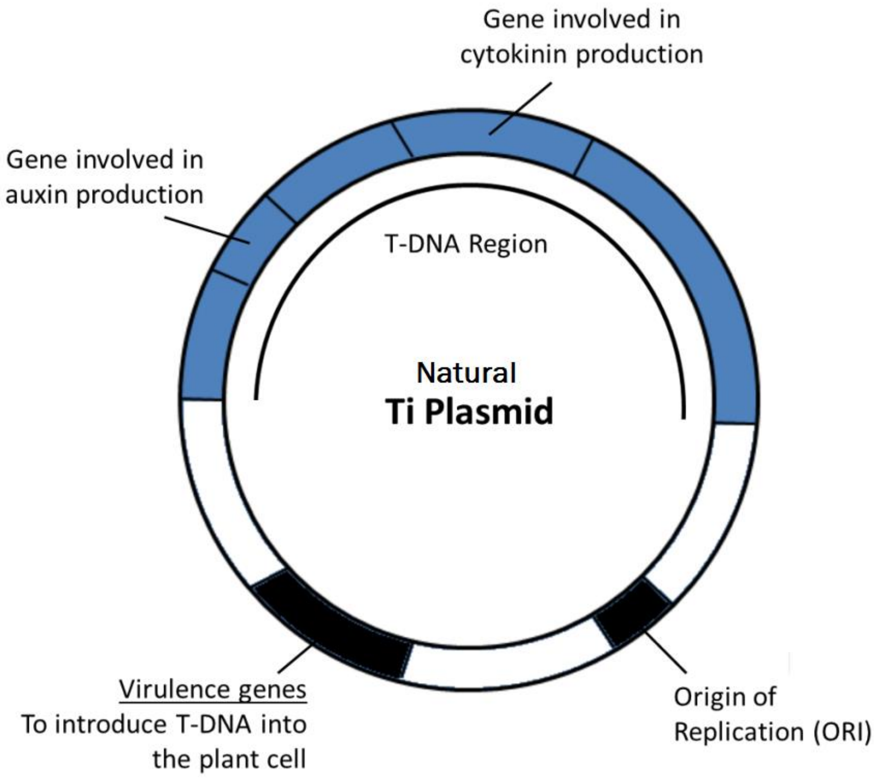

Figure 3.2

The genes for auxin and cytokinin are removed from the T-DNA region from the Ti plasmid before it can be used in the production of transgenic rice.

(a) Explain why the genes for auxin and cytokinin are removed from the T-DNA region of the Ti Plasmid.

Auxins and cytokinins are plant growth hormones;
These genes are expressed at high levels within the T-DNA region to promote tumour formation;
Tumours will depress the yield of transgenic rice that can be harvested;
Max 2

[2]

 [Turn over

For Examiner's Use

Transgenic rice plants have to undergo a process of acclimatization before they can be transferred to a natural environment for growth.

(b)

(i) Explain the importance of acclimatization.

Process of acclimatization allows plantlets to <u>gradually get used to external uncontrolled conditions</u> compared in vitro environment where they were previously grown in;
Plantlets are <u>chemoheterotrophs</u> and require <u>time to become photosynthetically competent</u>;
Max 1

(ii) Describe a key step in this process.

Plantlets are first <u>transplanted to sterile soil</u> for further growth in a <u>sheltered environment</u> before they are replanted in an open uncontrolled environment.
Max 1

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A study was carried out to evaluate the efficacy of such transgenic rice with increased iron content. Twenty iron-deficient 12-year old girls were selected for the study. The haemoglobin count for each girl chosen was similarly below 12.8gdl^{-3}. The normal haemoglobin count for girls at the age of 12 is 12.8 to 16.0gdl^{-3}.

The girls selected for the study were randomly and equally sorted into a treatment group and control group. The girls in the treatment group were requested to adhere to a diet of transgenic rice with increased iron content, three times a day, for 5 months. The girls in the control group were requested to adhere to a diet of normal rice grown in the same rice field, three times a day, for 5 months. The mean haemoglobin count for each group was tracked over this period.

The results of the analysis are shown in Figure 3.3.

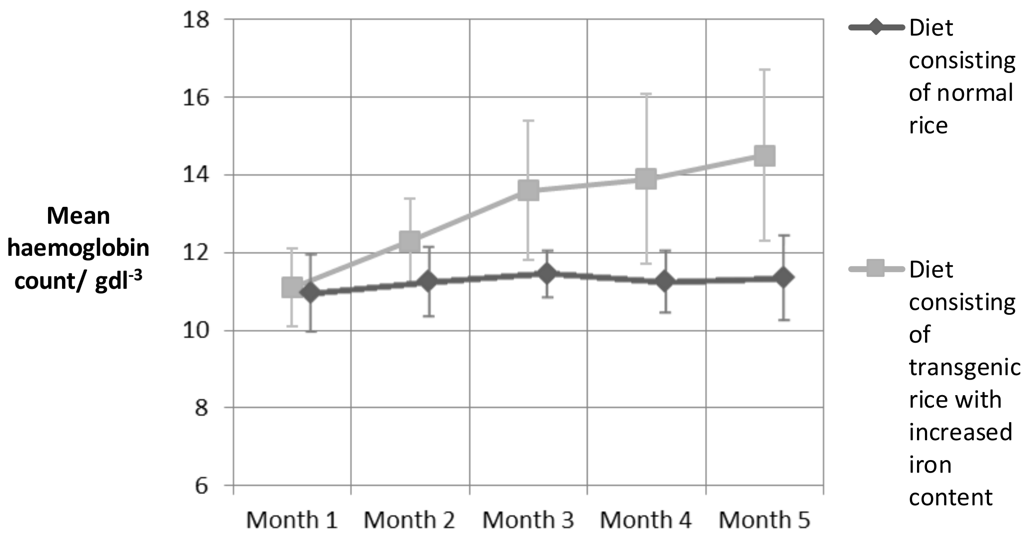


Figure 3.3

At the end of the study, the scientists involved concluded that the transgenic rice was more effective than normal rice in helping iron-deficient children.

(c) With reference to Figure 3.3, discuss the validity of this conclusion.

The conclusion is not valid;
The error bars for the mean haemoglobin count from transgenic rice diet indicate there is large/wide variation among the subjects studied;
This is exemplified by 11.5 to 16 gdl^{-3} for 4th month and 12.5 to 17 gdl^{-3} for 5th month
Therefore even through there seems a rise in mean haemoglobin count from transgenic diet compared to normal rice diet the difference in mean haemoglobin count between both diets is not considered significant;
The sample size/ the number of people studied is also too small to justify this conclusion;
Max 3m

For Examiner's Use

In addition to producing more nutritious food, genetically engineered crops like Bt corn can also be used in solving the world demand for food.

(d) Explain how Bt corn helps solve the global food challenge.

Bt gene codes for a large crystal-like protein known as the Bt toxin;
The toxin is ingested by target insect pests when they feed on transgenic corn;
The toxin binds to the specific receptors on the cell membranes of insect gut epithelial cells and causes them to be permeable;
The toxin therefore causes gut cells of insect to lyse, eventually leading to the death of the insects;
Bt toxin is not known to have any harmful effects on humans as the acid in our stomach will denature the protein;
Bt crop therefore increases yield of crops by reducing damage caused by insect pests;
Max4

[4]

[Total: 11]

4. Beetroot is a starchy edible root from the *Beta vulgaris* plant whose cells contain a water-soluble red pigment, betacyanin, in their vacuoles. Beet root pigment is used commercially as a food dye and different methods have been used to extract the pigment from intact beetroot cells. This pigment cannot pass through membranes unless the membranes are damaged.

A colorimeter can be used to measure the absorbance of light at 550 nm by betacyanin solution. The concentration of betacyanin is proportional to its absorbance. The use of a colorimeter to measure absorbance by betacyanin is depicted in Figure 4.1.

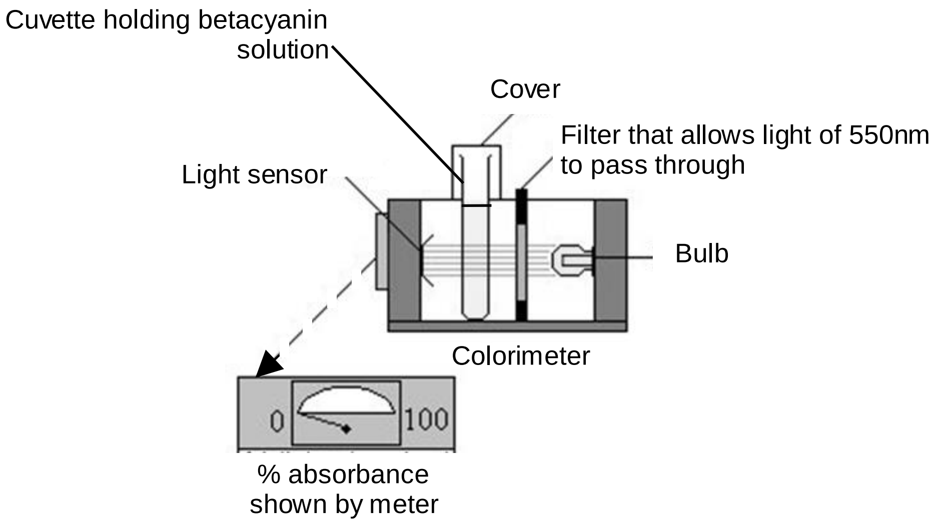


Fig. 4.1

Using the above information and your own knowledge, you are required to plan, but not carry out, an investigation into the effect of alcohol concentration on the permeability of the cell membrane of beetroot tissue.

Your planning must be based on the assumption that you have been provided with the following equipment and material which you must use.

- A large piece of beetroot tissue with the skin removed
- Sharp knife
- 10.0% Alcohol
- White tile
- Ruler
- Stopwatch
- Distilled water
- Colorimeter set at 550nm
- Cuvettes
- Normal laboratory glassware

Your plan should:

- have a clear and helpful structure such that the method you use can be repeated by anyone reading it,
- be illustrated by labelled diagrams, if necessary,
- identify independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results table(s) and graph(s) with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

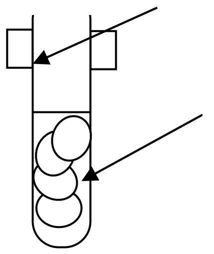
[Total: 12]

Section	Mark Scheme	Max marks
T	T1.Betacyanin pigment is large/ hydrophilic and cannot pass through hydrophobic core of phospholipid bilayer; T2.Increasing alcohol concentration increases damage/disruption to the integrity of cell membrane/regular arrangement of the membrane; T3.This is due to solubilisation of phospholipids/alcohol dissolving phospholipids; T4.Vacuole membrane/tonoplast AND cell surface membrane lose selective permeability/become more permeable, allowing pigment to diffuse out of the cell; T5.As concentration of alcohol increases, membrane permeability of beetroot tissue increases, leading to an increase in absorbance value;	Max 2 marks
	T6.The independent variable in this experiment is alcohol concentration while the dependent variable is absorbance at 550nm; T7.Other variables to be kept constant (any 2); -washing and rinsing cut beetroot cylinders before start of experiment. -calibration of colorimeter using alcohol solution; - Dimensions of beetroot tissue used (if using scalpel). -Time the beetroot cubes were immersed in the alcohol solution. -Same volume of alcohol used (e.g. 10-15cm^3) -Same cuvette used -Same beetroot used -Constant temperature (e.g. 28°C -30°C)	Max 1 mark
M	1. Prepare 10 c^{m3} each of 2.0%, 4.0%, 6.0%, 8.0% and 10.0% alcohol solution and control by diluting the 10.0% stock alcohol solution and place them in labelled test tubes;	6 marks max

Test tubes	Concentration of alcohol solution/%	Volume of 10% alcohol/cm^3	Volume of distilled water/cm^3
A	2.0	2.0	8.0
B	4.0	4.0	6.0
C	6.0	6.0	4.0
D	8.0	8.0	2.0
E	10.0	10.0	0.0
Control	0.0	0.0	10.0

1m for completed dilution table;

2. Cut the beetroot tissue into cubes of 1cm^3 (Or identical dimensions) on a white tile using a ruler and scalpel;
3. Wash and rinse beetroot cubes in distilled water to remove any pigments which may leak out during cutting. There should be no further leakage of colour from the beetroot tissue;
4. Place 5 beetroot cubes into a test tube A;
5. Immediately start timing for 5 minutes using a stopwatch;
6. After 5 minutes, remove 1 cm^3 of extract from test tube A and transfer to a cuvette;
7. Measure absorbance value at 550nm using colorimeter;
8. Repeat steps 4-7 for 4%, 6%, 8%, 10% alcohol solutions and control;
9. Before measuring absorbance, colorimeter should be calibrated with matching concentration of alcohol solution with respect to each sample;
10. Perform 2 more replicates for each concentration of alcohol solution to obtain average absorbance + repeat entire experiment twice with fresh batch of reagents to ensure reproducibility;



1m for labelled diagram;

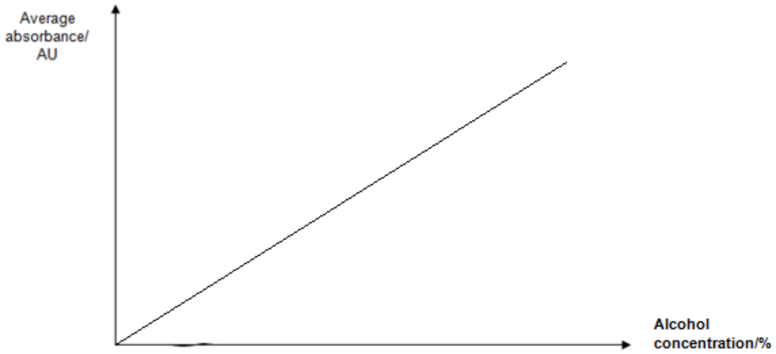
R

R1. Table with correct headings with appropriate units + Replicate and average included;

2 marks max

Concentration of alcohol solution/%	Absorbance / A.U			
	Replicate 1	Replicate 2	Replicate 3	Average
2.0				
4.0				
6.0				
8.0				
10.0				
0.0				

R2. Graph showing x-axis: Alcohol concentration A, y-axis: average absorbance / A.U;

		
S	S1.Handle scalpel carefully or place them away from main work area after use to avoid injuring/cutting oneself; S2.Handle liquid chemicals such as the 10% alcohol with care or wear protective goggles to prevent eyes from being in contact with alcohol; S3.Handle fragile objects like test tubes with care. Arrange test tubes neatly in the test tube rack, ensuring that they are placed where they cannot be knocked over or roll off the bench to prevent breakage;	1 mark max

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5.

(a) Describe cystic fibrosis and explain how it can be treated using a non-viral gene delivery system. [8]

(b) Discuss the advantages and limitations of plant tissue culture. [6]

(c) Discuss the ethical concerns pertaining to the use of genetically modified animals. [6]

[Total: 20]

End of Paper

For Examiner's Use

1.

(a) Describe cystic fibrosis and how it can be treated using a non-viral gene delivery system [8]

Description of cystic fibrosis:

Cystic fibrosis (CF) is an autosomal recessive disease;
Caused by mutation of cystic fibrosis transmembrane conductance regulator gene;
The CFTR protein is a chloride channel which regulates the transport of chloride ions across epithelial cells;
Mutations in the CFTR gene disrupt the function of the chloride channels, preventing them from regulating the flow of chloride ions and water across cell membranes;
The most common mutation is a deletion of three nucleotides on chromosome 7 that results in a loss of the amino acid phenylalanine (F) at the 508th (508) position on the protein;
As a result, cells that line the passageways of the lungs, pancreas, and other organs produce mucus that is unusually thick and sticky;
This abnormal mucus can clog the airways, leading to severe problems with breathing and bacterial infections in the lungs;
Over time, mucus build-up and infections result in permanent lung damage, including the formation of scar tissue (fibrosis) and cysts in the lungs;
A build-up of thick, sticky mucus in the pancreas can block the pancreatic duct from secreting digestive enzymes and importance hormones (e.g. insulin);
Problems with digestion can lead to diarrhoea, malnutrition, poor growth, and weight loss;
AVP
Max 4

Treatment of cystic fibrosis:

A normal functioning CFTR gene is isolated;
The allele is cloned many times by PCR;
The allele is then encapsulated into liposomes;
Liposomes are then delivered to epithelial cell in the lung by aerosol spray (nasal spray);
The gene particles are inhaled, so that they can pass into the epithelial cells of the lung to be incorporated into the DNA of the cells;
The liposome fuses with the plasma membrane of the epithelial cell lining the lung and releases the DNA into the cell;
The normal CFTR enters nucleus and is transcribed and translated to produce normal CFTR protein which can be incorporated into plasma membrane to facilitate the transport of chloride;
AVP
Max 4

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(b) Discuss the advantages and limitations of plant tissue culture. [6]

Advantages

Rate of growth by micro-propagation is also greater than conventional propagation;

The genetic makeup of all plants possessing desirable phenotypes can be preserved generation after generation;

Possible to produce clones of some plants species that are otherwise difficult or slow to propagate by conventional means (e.g. orchid);

Able to multiply plants which produce little or no seeds (e.g. bananas);

Production of disease free plants when explants are taken from meristematic regions (shoot tips, root tips), which are free from viruses;

Production of new plants can be continued all year round and is independent of seasonal/climate changes;

Relatively low space requirement, as compared to growing plants in greenhouses or farms;

Max 3

Disadvantages

The limited gene pool and genetic uniformity of plants cultured make them vulnerable to new diseases or drastic changes in the environment;

Very expensive due to several factors due to high overhead (specialized equipment, facilities, supplies) and labour costs;

It is also labour intensive as labour is required to transfer plantlets from laboratory to soil. Therefore this may not be economical for crops with low financial returns like carrots;

Plants produced from calli may undergo genetic changes to produce genetic variations, which usually result in undesirable phenotypes;

Max 3

(c) Discuss the ethical concerns pertaining to the use of genetically modified animals. [6]

Genetically modifying organisms will mean tampering with nature and hence going against the natural way of life;

There is concern about whether the animals are biologically capable of withstanding the additional stress of increased production of milk, meat and other products;

Increased use of growth hormone has harmful effects on the health of animals;

The use of bovine somatotropin in dairy cattle increases the risk of mastitis, which is a disease of the udder;

Medical experiments may cause suffering in animals;

For example, oncogenes are cloned into mice to study cancer. These transgenic mice develop tumours more frequently than normal ones and suffer the symptoms of cancer development;

In US, there is no mandatory labelling of GM food or GM processed food;

This may pose health risks as the consumer may have an unwanted reaction towards these products;

Some religions and ethnic groups avoid eating certain foods, but the food they eat might contain genes from other sources which they must not eat;

The concern rises from the belief that consumers should have right to know and choose what they are purchasing and consuming;

There is fear that cloning techniques (e.g. those used in the cloning of Dolly the sheep in 1997) may be used on humans, to create beings with desirable qualities race/ used to eventually practice eugenics;

[Total: 20]